Course Code : ECT 355-5 **GVHW/RW – 23 / 1220**

Fifth Semester B. Tech. (Electronics and Communication Engineering) Examination

DATABASE MANAGEMENT SYSTEMS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

All questions are compulsory.

1. (a) Enlist and explain various abstraction levels of database. 5(CO1)

(b) Consider the two tables T1 and T2 –

Table T1

P	Q	R
10	a	5
15	b	8
25	a	6

Table T2

A	B	C
10	b	6
25	c	3
10	b	5

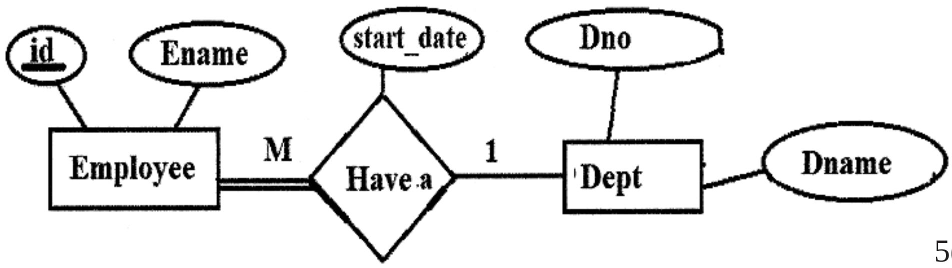
Evaluate the results of following operations :

(i) $T1 \bowtie_{T1 \cdot P = T2 \cdot A} T2$.

(ii) $T1 \bowtie_{T1 \cdot P = T2 \cdot A} T2$.

(iii) $T1 \bowtie_{(T1 \cdot P = T2 \cdot A \text{ AND } T1 \cdot R = T2 \cdot C)} T2$. 5(CO1)

2. (a) Convert the ER model into Relation(s).



5(CO2)

(b) For the given relation R(A, B, C, D, E) with Functional Dependencies : $\{AB \rightarrow CD,\ D \rightarrow A,\ BC \rightarrow DE\}$. Identify the given relation, R in which normal form. If the relation R is not in BCNF then convert it into BCNF. 5(CO1)

3. (a) Enlist aggregate operators in SQL. Write a SQL query for ANY TWO aggregate operators for the following relation PRODUCT_MASTER.

PRODUCT_MASTER

PRODUCT	COMPANY	QTY	RATE	COST
Item1	Com1	2	10	20
Item2	Com2	3	25	75
Item3	Com1	2	30	60
Item4	Com3	5	10	50
Item5	Com2	2	20	40

5(CO3)

(b) Design a database considering the following database schema –

Employee (Ename, Street, City)

Works (Ename, Company_name, Salary)

Company (Company_name, C_city)

Write SQL queries for the following :

(i) Retrieve name of all employees who works for the company "Google".

(ii) Find the name of the person(s) who live in the same city and same street as that of "Mr. Sonraj". 5(CO3,5)

4. (a) What are the differences among primary, secondary and clustering indexing ? How to these differences affect the ways in which these indexes are implemented ? Which of the indexes are dense and which are not ? 5(CO2)

(b) Consider a hash table of size seven, with starting index zero, and a hash function, $h(K) = (3K + 4)$. Assuming the hash table is initially empty, determine the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing ? 5(CO2)

5. (a) Write a short note on Query Optimization. 5(CO2)

(b) Consider doing a natural join operation on two relations R and S. Assume that the 5000 tuples of R are stored contiguously on 100 disk blocks, while the 10000 tuples of S are stored contiguously on 400 blocks. Assume that relation R is an outer relation and relation S is an inner relation.

Calculate number of block transfers, number of seeks and cost of the query for both Worst case and Best case. 5(CO2)

6. (a) Consider the following schedules :

$$S_1 : r_1(x),\ r_1(y),\ r_2(x),\ r_2(y),\ w_2(y),\ w_1(x)$$

Check whether schedule S_1 is a conflict serializable or not. If not, then convert it into conflict serializable schedule. 5(CO4)

(b) Which are the modes of lock ? Explain two phase locking protocol. 5(CO4)